Distributed Edge Artificial Intelligence

A Thesis (Major Project) submitted in partial fulfilment of the requirements

For the Degree of

Bachelor of Technology

In

Computer Science and Engineering

Submitted by

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Declaration

We, the students of Computer Science & Engineering, Barak Valley Engineering College, declare that the project report entitled **"Distributed Edge Artificial Intelligence"** has been prepared under the guidance of **Mr. Rana Singha**, Computer Science & Engineering, Barak Valley Engineering College.

The report has been prepared without resorting to plagiarism. We have adhered to the principles of academic honesty and integrity. No falsified or fabricated data have been presented in this report. Further, the matter embodied in this project report has not been submitted previously for the award of any degree or diploma.

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Abstract

Distributed Edge Artificial Intelligence (Edge AI) places artificial intelligence workloads across devices near data sources, edge servers, and cloud resources. This report presents a preliminary literature study of how such workloads can be distributed while accounting for communication, computing capacity, energy use, and privacy. A thematic review is used to examine end-edge-cloud architectures, federated learning, collaborative inference, communication efficiency, resource heterogeneity, and security considerations. The reviewed work indicates that distributing computation closer to data can support responsive applications and reduce dependence on centralised processing, but the benefits depend on task placement, network conditions, and device capabilities. Federated training and partitioned inference are identified as complementary approaches, each with trade-offs in communication, coordination, and resource requirements. The project remains at the literature-review and feasibility-assessment stage; no implementation or experimental measurements are claimed. Future work will select an application and platform, then evaluate an appropriate distributed-AI design using reproducible performance and energy measures.

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Chapter 1

Introduction

Artificial intelligence applications increasingly need to process information close to where it is produced. Distributed Edge Artificial Intelligence studies how AI workloads can be placed across resource- constrained devices, nearby edge servers, and cloud infrastructure. This approach can reduce reliance on centralised processing, but it also raises challenges involving communication, available computing resources, energy, and coordination.

1.1 Background

An edge-based system may include sensors, user devices, gateways, and computing infrastructure. The placement of a task can affect response time, network traffic, privacy, and resource use. These considerations motivate studying the design choices involved in distributing AI workloads.

1.2 Problem Statement

This project examines the design choices involved in distributing AI workloads across edge resources. In particular, it considers how communication cost, device and data heterogeneity, energy constraints, and privacy requirements affect distributed training and inference. The current work is a literature review and feasibility assessment; a specific application and implementation platform remain to be selected.

1.3 Objectives

The objectives of this preliminary study are:

1. Review distributed edge-AI architectures and approaches to federated training and collaborative inference.

2. Identify key challenges related to communication, heterogeneity, privacy, energy, and available hardware resources.

3. Establish a feasibility-assessment framework to guide selection of an application and platform for future implementation and evaluation.

1.4 Scope and Report Organization

This report is organized into five chapters. Chapter 1 introduces the topic, problem, objectives, and scope. Chapter 2 reviews related work in distributed edge AI. Chapter 3 describes the literature-review protocol and proposed feasibility and technical approaches. Chapter 4 synthesizes the review findings and discusses feasibility and limitations. Chapter 5 presents the conclusions and proposed future work.

Chapter 2

Literature Survey

2.1 From Cloud AI to Edge Intelligence

The reviewed literature positions distributed edge AI as a response to the limits of centralised cloud processing. Edge AI performs AI computation near the user at the network edge rather than in a centralised data centre; this shift has been enabled by improvements in AI efficiency, the proliferation of IoT devices, and the growth of edge computing. Zhou et al. describe edge intelligence as extending AI to the "last mile" for mobile and IoT devices that generate data at the edge. Deng et al. distinguish AI for edge from AI on edge, with the latter covering model training and inference on edge devices [2]. The term "edge" has no universally accepted definition; its relation to cloud computing depends on proximity, capacity, and device heterogeneity [2].

2.2 End-Edge-Cloud Architectures

A common architectural pattern distributes processing across end devices, edge servers, and the cloud. End devices handle lightweight local computation, while edge servers provide nearby, more capable resources. The literature distinguishes the complementary roles of AI for edge and AI on edge, and describes orchestration of heterogeneous on-device, edge, and cloud resources for distributed AI. Collaborative learning across cloud and edge resources is also studied [2]. Hu et al. describe a distributed edge-AI scheduler that places application tasks across heterogeneous nodes while accounting for changing system conditions [1].

2.3 Distributed Training: Federated Learning

Federated learning (FL) trains local models on multiple clients without collecting raw user data at a central data centre. Federated learning and knowledge distillation are described as training approaches, while model splitting and partitioning support inference. FL challenges include statistical heterogeneity, communication cost, resource allocation, and privacy and security. Communication can dominate computation when local datasets are small and participant processors are comparatively fast [2].

Communication efficiency, resource allocation, and data heterogeneity are recurring concerns in distributed edge learning [2].

2.4 Distributed Inference: Partitioning and Collaboration

Model-partition methods split a deep neural network between a device and server based on transmission latency and device energy. Early-exit methods can reduce communication when intermediate features are sufficient. DistrEdge applies deep reinforcement learning to distribute CNN inference across heterogeneous IoT edge devices and reports a 1.1–3× speedup over the compared methods [3]. Edge-intelligence surveys also cover the complementary roles of distributed training and inference [2].

2.5 Communication Efficiency

Communication is a central bottleneck across training and inference. What is transmitted affects per-round overhead and the number of communication rounds. Quantisation and pruning, together with in-network computation, are discussed as ways to address communication cost. Computation-intensive inference may require task splitting across edge devices and servers, making communication efficiency especially important [2, 3].

2.6 Heterogeneity, Non-IID Data, and Stragglers

FL over edge nodes must address device heterogeneity, non-independent and identically distributed (non-IID) data, and limited communication resources. Synchronous FL can wait for the slowest worker, while asynchronous FL may cause frequent model transfers. The straggler effect can limit synchronous training time. Non-IID and imbalanced data can cause local updates to drift, resulting in instability and slower convergence; system heterogeneity can also lead to stragglers and dropouts. Participant selection is one proposed mitigation, while edge-assisted architectures, data augmentation, and mediator-based rescheduling are reported as approaches to non-IID bias. These challenges are part of the broader edge-intelligence research landscape [2].

2.7 Privacy and Security

FL does not eliminate privacy risks: a malicious participant may infer sensitive attributes from shared models, and systems may be vulnerable to data or model poisoning. One example in the reviewed literature reports inferring whether an individual's input appeared in a gender-classifier training set with up to 90% accuracy. Other surveyed threats include honest-but-curious servers, free-riding participants, poisoning, and model inversion. Inference attacks may recover training samples; differential privacy is one mitigation, with

a possible trade-off against model performance. Surveys classify poisoning and inference attacks and identify communication blockage, poisoning, and backdoor attacks among high-priority threats. Privacy-preserving cloud-edge-end architectures include encrypted or anonymised protocols and multi-centre hierarchical FL. Blockchain-based FL is also discussed as a way to remove a central aggregator.

2.8 Energy and Hardware Constraints

Latency, energy efficiency, and computation or memory footprint are important evaluation metrics; latency-sensitive applications may have constraints around 100 ms. Energy-efficient edge training is another motivation in the literature; one cited example uses AlphaGo's reported requirement of 280 GPUs and a $3,000 electricity cost per game to illustrate the cost of computation. Reported Raspberry Pi 5 measurements for MobileNetV3-Small include 2.14–2.17 ms latency, approximately 2.55 W net power, and 5.5 mJ per inference. Reported ResNet-18 measurements include 12.9–14.1 ms, 2.4–2.8 W, and 31–39 mJ per inference.

Other reported results include up to 10 W for a Jetson Nano GPU configuration, approximately 5 W for a Google Coral TPU, and approximately 2 W for an Intel NCS2 accelerator. Reported power values for an Arduino Nano, Raspberry Pi, and idle Jetson Nano are 0.17 W, 3.5 W, and 1.3–1.6 W, respectively. Another comparison reports a Jetson Nano approximately three times faster and 1.67 times lower in energy than a Raspberry Pi for its network configuration. These results use different platforms, workloads, and protocols and are indicative rather than directly comparable.

2.9 Application Domains

Reviewed applications include autonomous and connected vehicles, smart homes, industrial automation, healthcare, surveillance, real-time gaming, smart factories, and precision-agriculture pest detection.

Chapter 3

Methodology

Because the project remains at the learning and literature-review stage and the application is undecided, the methodology below is proposed rather than executed.

3.1 Review Protocol

The preliminary study follows a structured literature-review approach. Research questions are modelled on existing reviews of computing, storage, and communication technologies in distributed edge AI and optimisation along the edge-fog-cloud data path. The approach also draws on systematic reviews that develop taxonomies of edge AI systems and examine cloud-edge AI integration [2, 1]. Sources are selected from academic databases for relevance to architecture, training, inference, communication, heterogeneity, privacy and security, energy, and applications.

3.2 Analytical Framework

Sources are synthesised across eight themes: definition and motivation; architecture; training; inference; communication; heterogeneity; privacy and security; and energy and hardware. This structure is intended to help map a later application choice to the relevant literature.

3.3 Proposed Feasibility-Assessment Framework

A multi-criteria assessment is proposed using latency, energy efficiency, and computation and memory footprint, extended with bandwidth sensitivity, scalability, and cost. Candidate platforms could be compared using a reproducible benchmarking approach that measures latency, throughput, power, and energy.

3.4 Proposed Technical Methodology

Two possible implementation directions are proposed for a later phase:

1. **Partitioned collaborative inference:** split a model by layer or horizontally across devices and exchange intermediate outputs, following published partitioning schemes [3].

2. **Federated training:** participants train local models and share model updates through an aggregation server, as surveyed for mobile edge networks and edge training [2].

Neither direction has been implemented; both remain proposals for subsequent work.

Chapter 4

Results and Discussion

As this is a preliminary, literature-stage study, the results are the outcomes of the review and feasibility synthesis rather than experimental findings.

4.1 Literature Synthesis

The reviewed literature converges on layered end-edge-cloud architectures, with federated training and partitioned or collaborative inference as two major technical families [2, 3]. Communication efficiency is a recurring bottleneck across both.

4.2 Feasibility Indications

The literature suggests that distributed edge AI can support latency- sensitive and privacy-sensitive workloads, while remaining constrained by communication and energy. In one reported setup, edge-server inference increased from 0.123 seconds at 1 Mbps to 2.317 seconds at 50 kbps, while a mobile device took more than 2 seconds. Hardware energy figures vary substantially by device and workload.

Table 4.1: Hardware metrics reported in the reviewed literature.

Platform or workload	Metric	Reported value
Raspberry Pi 5, MobileNetV3-Small	Latency, net power, energy	2.14–2.17 ms; about 2.55 W; 5.5 mJ per inference
Raspberry Pi 5, ResNet-18	Latency, net power, energy	12.9–14.1 ms; 2.4–2.8 W; 31–39 mJ per inference
NVIDIA Jetson Nano	Peak power	Up to 10 W
Google Coral TPU	Power	About 5 W
Intel NCS2 and Raspberry Pi 3B+	Accelerator and host power	About 2 W and 3 W
Arduino Nano	Power	0.17 W
Raspberry Pi	Power	3.5 W
NVIDIA Jetson Nano	Idle power	1.3–1.6 W

Partitioned collaborative inference is reported to achieve 1.1–3× speedups over the methods compared in DistrEdge [3]. A communication- efficient FL method reports up to 87% improvement in convergence time and up to 3.6% improvement in accuracy relative

to standard FL. These results come from different hardware, workloads, and measurement protocols and should not be compared directly.

4.3 Discussion and Limitations

The synthesis indicates a trade-off between latency and energy on one side and communication cost on the other. Moving computation closer to the data may reduce latency and bandwidth pressure, but lower edge computing capacity and device heterogeneity can require model partitioning, compression, or collaborative execution. Non-IID data and slow or unreliable nodes can create stragglers and dropouts. Privacy and security risks, including inference and poisoning attacks, also constrain how data and model updates are handled.

No implementation or measurements have been carried out; this feasibility assessment relies entirely on published results. Those results use different platforms, workloads, and measurement setups. Even power-measurement protocols vary; one study reports reducing measurement uncertainty by factors of 6 and 12 on two device families relative to an established protocol. Direct cross-study comparison of energy values must therefore be treated cautiously. The field is still developing, and the definition of "edge" remains unsettled. Open challenges include resource constraints, security threats, scalability, and communication overhead.

Chapter 5

Conclusion and Future Work

This preliminary report has reviewed distributed edge AI and assessed feasibility at a conceptual level. The literature describes a shift from centralised cloud AI toward layered end-edge-cloud architectures, with federated learning and collaborative DNN partitioning as major technical families. Potential suitability for latency- and privacy-sensitive applications is bounded by communication cost, device and data heterogeneity, energy constraints, and privacy and security risks [2, 3].

Because the application and implementation remain undecided, the next phase is to select a specific application and platform using the proposed review protocol and feasibility framework, then design an implementation for future evaluation. No implementation or experimental result is claimed here.

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The links above lead to the online publication records. Local copies are stored in the `references/` folder.